25901579 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCZGY201 - BIOLOGICAL CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. A complete, catalytically active enzyme consisting of apoenzyme and cofactor is known as: [K] / [CO2]

   a). Apoenzyme b). Abzyme

   c). Holoenzyme d). Isoenzyme

2. Fructose-1,6-bisphosphatase converts fructose-1,6-bisphosphate to: [K] / [CO4]

   a). Fructose-6-phosphate b). Glucose-6-phosphate

   c). Fructose-1-phosphate d). Glucose-1-phosphate

3. Carbohydrates containing a ketone group are called: [K] / [CO1]

   a). Aldoses b). Pentoses

   c). Hexoses d). Ketoses

4. Anaerobic respiration in yeast produces---------- [K] / [CO4]

a). Lactic acid and CO_2 b). Ethanol and CO_2

c). Pyruvic acid and ATP d). Acetic acid and O_2

5. Which of the following amino acids is purely ketogenic? [K] / [CO1]

a). Valine b). Leucine

c). Alanine d). Glycine

6. Cholesterol is delivered to peripheral tissues by--------- [K] / [CO4]

a). LDL b). VLDL

c). HDL d). None of the above

7. Which of the following is a function of phospholipids? [K] / [CO1]

a). Enzyme synthesis b). Energy storage

c). Formation of biological membranes d). Oxygen transport

8. Which molecule acts as an allosteric activator of PFK-1? [K] / [CO2]

a). ATP b). Citrate

c). AMP d). NADH

9. Starch is digested by salivary and pancreatic amylase into ------ [K] / [CO1]

a). Fructose b). Maltose

c). Glycogen d). Lactose

10. Bee wax is an ester of …… and a long chain ……. [K] / [CO1]

a). Phospholipids, alcohol b). Steroids, glycerol

c). Palmitic acid, alcohol d). Glycolipids, alcohol

11. During glycogenolysis, glycogen phosphorylase cleaves glycogen to produce: [K] / [CO4]

a). Free glucose b). Glucose-6-phosphate

c). Glucose-1-phosphate d). Fructose-6-phosphate

12. NADH stores energy in the form of …….. [K] / [CO3]

a). Phosphate bonds b). Electrons

c). Protons d). Heat

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. Mention the biological functions of RNA. [U] / [CO1]

14. List the major risk factors for atherosclerosis. [U] / [CO4]

15. Explain the activation energy in enzymology. [U] / [CO2]

16. What is oxidative deamination? Name the enzyme involved. [U] / [CO4]

17. Explain the importance of NAD^+ in glycolysis and fermentation. [U] / [CO4]

18. Explain DNA enzymes (Deoxyribozymes). [U] / [CO2]

19. Describe exergonic reactions. [U] / [CO3]

20. Describe the structure of glucose. [U] / [CO1]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21. Discuss how enzymes increase the rate of a chemical reaction. [U] / [CO2]

22. Write an account on molecular chaperones and their biological importance. [U] / [CO1]

23. Describe β-oxidation of saturated fatty acids. [U] / [CO4]

24. Summarize the importance of polysaccharides and stereoisomers. [U] / [CO1]

25. Explain uncompetitive inhibition and describe how it affects enzyme–substrate complex formation. [U] / [CO2]

26. Describe the overall energy yield during the oxidation of glucose. [U] / [CO4]

[5x4 = 20]